



Advanced Time Series Analysis and Forecasting

Seminar 1: Forecast evaluation, scoring rules and combination



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARX: AutoRegressive model with eXogenous variables
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID: COronaVirus Disease (2019 pandemic)
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ DM: Diebold–Mariano (test)
- ▣ ENTSO-E: European Network of Transmission System Operators for Electricity
- ▣ ES: Expected Shortfall
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ GW: Giacomini–White (test of conditional predictive ability)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HICP: Harmonised Index of Consumer Prices
- ▣ HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- ▣ IS: In-Sample
- ▣ LR: Likelihood Ratio (test)
- ▣ MA: Moving Average
- ▣ MAE: Mean Absolute Error
- ▣ MCS: Model Confidence Set
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ PIT: Probability Integral Transform
- ▣ RMSE: Root Mean Squared Error
- ▣ RTDSM: Real-Time Data Set for Macroeconomists (Philadelphia Fed)

Acronyms used in this material (2/2)

- ▣ SCA: Seasonally and Calendar Adjusted
- ▣ SE: Standard Error
- ▣ SPA: Superior Predictive Ability (test)
- ▣ SPF: Survey of Professional Forecasters (Federal Reserve Bank of Philadelphia)
- ▣ UNEMP4: SPF variable: unemployment rate forecast for the quarter two quarters after the survey
- ▣ UNRATE: Unemployment Rate (FRED series code)
- ▣ US: United States

Today's question and route

- **Question:** how do we decide, with the right inference, that one forecast is better than another, and when does averaging forecasts help?
 - ▶ this seminar comes **before** Lecture 1: the section “Prerequisites for Today” gives every definition the tasks use
- **Route**
 - ▶ Part A: optimal forecasts under four losses; log score, CRPS, an improper score, the interval score; DM, HLN and Clark–West by hand; Bates–Granger weights and the cost of estimating them
 - ▶ Part B: Romanian GDP growth; SPF unemployment forecasts; EUR/RON density forecasts; GDP data vintages; Romanian electricity load at the evening peak
 - ▶ Part C: a pre-registered evaluation of Romanian inflation forecasts by horizon; an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section

Exercise map

Task	Question	Type	Model
A1, A2	optimal point forecasts: squared, absolute, pinball, LinEx	Solved, Proposed	A1
A3, A4	log score and CRPS; an improper score; the interval score	Solved, Proposed	A3
A5, A6	DM and HLN by hand; Clark–West by hand	Solved, Proposed	A5
A7, A8	Bates–Granger weights; the cost of estimated weights	Solved, Proposed	A7
B1, B2	Romanian GDP growth; combining SPF unemployment forecasts	Solved, Proposed	B1
B3–B5	EUR/RON densities; GDP vintages; load quantiles at the evening peak	Solved, Proposed	B1, B3
C1, C2	inflation forecasts by horizon; what is wrong in an AI answer?	Proposed	B1, A1–A8

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
real GDP, Romania (SCA)	Eurostat (namq_10_gdp)	quarterly	1995–2026
SPF individual forecasts	Philadelphia Fed	quarterly	1990–2025
unemployment rate, US	FRED (UNRATE)	monthly	1990–2026
EUR/RON	BNR reference rate	daily	2010–2026
real GDP growth, all releases, US	Philadelphia Fed (RTDSM)	quarterly	1990–2025
electricity load, Romania	Energy-Charts (ENTSO-E data)	hourly	2023–2026
HICP inflation, Romania	Eurostat (prc_hicp_minr)	monthly	1997–2026

- In the notebook: `dm_test`, `cw_test`, `mz_test`, `berkowitz_test`, `crps_normal`, `combine_spf`, `load_forecasts`; no account or key is needed

Prerequisites for Today (1/6): losses and optimal forecasts

- Point forecast x , outcome y , forecast error $e = y - x$; $Y \sim F$ (the predictive distribution)
 - ▶ the **optimal** forecast minimises the expected loss $\mathbb{E}_F L(x, Y)$ over x
 - ▶ a loss is **consistent** for a functional (mean, quantile) if that functional minimises the expected loss for every F
- Three losses and their optimal forecasts
 - ▶ squared $e^2 \rightarrow$ the mean; absolute $|e| \rightarrow$ the median
 - ▶ **pinball** $\rho_\tau(e) = (\tau - 1\{e < 0\})e \rightarrow$ the τ -quantile $F^{-1}(\tau)$; first-order condition $F(x) - \tau = 0$
 - ▶ $\tau \in (0, 1)$: the quantile level; $1\{e < 0\}$: 1 if the forecast is too high, 0 otherwise

Prerequisites for Today (2/6): asymmetric loss

- ▣ **LinEx** $L(e) = b[e^{ae} - ae - 1]$, $b > 0$
 - ▶ a : the asymmetry; $a > 0$: under-forecasts ($e > 0$) cost exponentially, over-forecasts about linearly
 - ▶ optimal forecast $x^* = a^{-1} \ln \mathbb{E}e^{aY}$
- ▣ Moment generating function of $N(\mu, \sigma^2)$:
 - ▶ $\mathbb{E}e^{aY} = \exp(a\mu + a^2\sigma^2/2)$
 - ▶ $Y \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$: the outcome is Normal given the information $\mathcal{F}_t$ at the forecast origin

Prerequisites for Today (3/6): probabilistic forecasts and scores

- ▣ **PIT** (probability integral transform): $u_t = F_t(y_t)$, the forecast probability of an outcome below y_t
 - ▶ if F_t is the true one-step conditional distribution, u_t are i.i.d. $U(0, 1)$ [Diebold, Gunther and Tay \(1998\)](#)
 - ▶ U-shaped histogram: forecasts too sharp; hump: too wide; slope: biased
- ▣ **Berkowitz test** [Berkowitz \(2001\)](#): $z_t = \Phi^{-1}(u_t)$ should be i.i.d. $N(0, 1)$
 - ▶ Φ^{-1} : the standard Normal quantile function; fit an AR(1) to z_t with mean μ , variance σ^2 , autocorrelation ρ
 - ▶ LR (likelihood ratio) test of $\mu = 0$, $\sigma = 1$, $\rho = 0$, reference distribution $\chi^2(3)$

Prerequisites for Today (4/6): proper scores

- **Scores** (lower is better); **proper**: the true distribution minimises the expected score [Gneiting and Raftery \(2007\)](#)
 - ▶ log score $-\ln f(y)$, with f the forecast density
- **CRPS** of $N(\mu, \sigma^2)$, with $z = (y - \mu)/\sigma$:
 - ▶ $\sigma[z(2\Phi(z) - 1) + 2\varphi(z) - 1/\sqrt{\pi}]$
 - ▶ Φ, φ : the distribution and density functions of $N(0, 1)$; the CRPS is in the units of y
- **Interval score** of a central interval $[l, u]$ at level $1 - \alpha$:
 - ▶ $(u - l) + \frac{2}{\alpha}(l - y)1\{y < l\} + \frac{2}{\alpha}(y - u)1\{y > u\}$
 - ▶ the width plus a penalty of $2/\alpha$ per unit by which y falls outside

Prerequisites for Today (5/6): comparing forecasts

- **Diebold–Mariano** [Diebold and Mariano \(1995\)](#): loss differential $d_t = L(e_{1t}) - L(e_{2t})$ over P forecasts
 - ▶ $DM = \bar{d} / \sqrt{\hat{\sigma}_{LR}^2 / P}$, $\hat{\sigma}_{LR}^2 = \hat{\gamma}_0 + 2 \sum_{k=1}^{h-1} \hat{\gamma}_k$
 - ▶ $\bar{d}$: the mean differential; $\hat{\gamma}_k$: the sample autocovariances of d_t ; $\hat{\sigma}_{LR}^2$: its long-run (HAC) variance, [Chapter 0](#)
 - ▶ overlapping h -step targets make d_t an $MA(h-1)$: never use the i.i.d. variance for $h > 1$
- **HLN** [Harvey, Leybourne and Newbold \(1997\)](#): $DM^* = DM \sqrt{(P+1-2h+h(h-1)/P)/P}$, compared with t_{P-1}
 - ▶ the factor corrects the small-sample bias of $\hat{\sigma}_{LR}^2$; t_{P-1} : Student t with $P-1$ degrees of freedom
- **Nested** models (small model 1 inside model 2): **Clark–West** [Clark and West \(2007\)](#)
 - ▶ $f_t = e_{1t}^2 - [e_{2t}^2 - (\hat{y}_{1t} - \hat{y}_{2t})^2]$, one-sided t test of $\mathbb{E}f_t = 0$; $\hat{y}_{it}$: the forecast of model i
- **Mincer–Zarnowitz** [Mincer and Zarnowitz \(1969\)](#): regress $y_{t+h} = a + b\hat{y}_{t+h|t} + u_{t+h}$
 - ▶ efficiency means $(a, b) = (0, 1)$; Wald test with HAC errors

Prerequisites for Today (6/6): combination and real-time data

- **Bates–Granger** [Bates and Granger \(1969\)](#): errors with variances σ_1^2, σ_2^2 and covariance σ_{12} ; weight w on forecast 1
 - ▶ $w^* = (\sigma_2^2 - \sigma_{12})/D$, combined variance $(\sigma_1^2\sigma_2^2 - \sigma_{12}^2)/D$, $D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$
 - ▶ **combination puzzle**: equal weights often beat estimated ones, because estimation adds $\text{Var}(\hat{w})D$ to the MSE [Smith and Wallis \(2009\)](#)
 - ▶ schemes: mean, median, trimmed mean, inverse-MSE weights, shrinkage $\lambda\hat{w} + (1 - \lambda)/N$ (N forecasts), previous best [Genre et al. \(2013\)](#)
- **Real-time data**: each **vintage** is the data as published on a date; GDP is revised for years [Croushore and Stark \(2001\)](#)
 - ▶ SPF: the Survey of Professional Forecasters (Philadelphia Fed), quarterly since 1968, individual answers public
- **Holm** correction for m tests
 - ▶ order the p -values, multiply the k -th smallest by $m - k + 1$, keep them increasing

A1: optimal forecasts of an exponential variable [Solved]

Task

- ▣ $Y \sim \text{Exponential}(1)$: $F(y) = 1 - e^{-y}$, $\mathbb{E}Y = 1$, $\text{Var}Y = 1$.
- ▣ 1. Derive the optimal forecast under the squared, the absolute and the pinball loss with $\tau = 0.9$.
- ▣ 2. Give the minimum expected loss (the risk) under the squared and the absolute loss.
- ▣ 3. A forecaster reports the median and is judged by RMSE. Compute the excess expected squared loss.
- ▣ Report: three forecasts, two risks, one excess loss.

Solution

- ▣ 1. Squared: $x^* = \mathbb{E}Y = 1$; absolute: $F(x) = 1/2 \Rightarrow x^* = \ln 2 = 0.693$; pinball: $F(x) = 0.9 \Rightarrow x^* = \ln 10 = 2.303$
- ▣ 2. Squared risk = $\text{Var}Y = 1$; absolute risk $\mathbb{E}|Y - \ln 2| = \ln 2 = 0.693$ (by integration); pinball risk at $\ln 10$: 0.230
- ▣ 3. $\mathbb{E}(Y - m)^2 = \text{Var}Y + (1 - m)^2$, excess $(1 - \ln 2)^2 = 0.094$: the honest median looks 9% worse under MSE
- ▣ the loss must be announced before the forecast is made

A2: the optimal forecast under LinEx loss [Proposed]

Solution discussed in class

Task

- ▣ LinEx $L(e) = b[e^{ae} - ae - 1]$, $e = y - x$, $b > 0$; $Y \mid \mathcal{F}_t \sim N(\mu_t, \sigma_t^2)$. Model: A1.
- ▣ 1. Show that the first-order condition is $\mathbb{E}e^{a(Y-x)} = 1$ and solve it for x .
- ▣ 2. Compute x^* for $\mu = 2$, $\sigma = 1.5$ and $a = 0.5$, then $a = -0.5$.
- ▣ 3. Explain why a rational forecaster with this loss fails the Mincer–Zarnowitz test, and why the bias moves with σ_t .
- ▣ Report: one formula, two numbers, two sentences.

A3: log score and CRPS of two Normal forecasts [Solved]

Task

- ▣ Two forecasts of y : $N(2, 1)$ and the sharper $N(2, 0.5^2)$; the outcome is $y = 3.5$.
- ▣ 1. Compute z , the log score $-\ln f(y)$ and the CRPS of each forecast.
- ▣ 2. Which forecast wins under each score, and which score penalises the miss more?
- ▣ 3. Prove that the log score is proper using $\text{KL}(g\|f) \geq 0$.
- ▣ Report: a table of four scores, one sentence, a two-line proof.

Solution

- ▣ 1. $N(2, 1)$: $z = 1.5$, $\text{LogS} = \frac{1}{2} \ln 2\pi + z^2/2 = 2.044$, $\text{CRPS} = 0.994$
- ▣ $N(2, 0.25)$: $z = 3.0$, $\text{LogS} = 4.726$, $\text{CRPS} = 1.218$
- ▣ 2. The wider forecast wins under both; LogS more than doubles for the sharp one (z^2 grows fourfold), CRPS rises by about a fifth: the log score punishes overconfidence harder
- ▣ 3. $\mathbb{E}_g[-\ln f(Y)] - \mathbb{E}_g[-\ln g(Y)] = \int g \ln(g/f) = \text{KL}(g\|f) \geq 0$, with equality iff $f = g$: strictly proper

A4: an improper score and the interval score [Proposed]

Solution discussed in class

Task

- ▣ Truth $Y \sim N(0, 1)$; forecasts $N(0, s^2)$. The *linear score* is $-f(y)$. Model: A3.
- ▣ 1. Show that $\mathbb{E}f(Y) = [2\pi(1 + s^2)]^{-1/2}$ and evaluate it for $s = 0.5, 1, 2$.
- ▣ 2. Explain why the linear score is improper and what behaviour it rewards.
- ▣ 3. A 90% interval is $[1, 4]$. Compute the interval score for $y = 3$, $y = 5$ and $y = -1$.
- ▣ Report: three expectations, one sentence, three scores.

A5: the Diebold–Mariano test by hand [Solved]

Task

- ▣ Two-step forecasts ($h = 2$), $P = 40$; squared-loss differential with $\bar{d} = 0.30$, $\hat{\gamma}_0 = 1.2$, $\hat{\gamma}_1 = 0.4$.
- ▣ 1. Compute the long-run variance with the rectangular kernel and $h - 1$ lags, the standard error of $\bar{d}$ and the DM statistic.
- ▣ 2. Apply the HLN correction and compare with t_{39} .
- ▣ 3. What would the i.i.d. variance have concluded?
- ▣ Report: two statistics, two p -values, one sentence.

Solution

- ▣ 1. $\hat{\sigma}_{LR}^2 = 1.2 + 2 \cdot 0.4 = 2.0$; $SE = \sqrt{2/40} = 0.224$; $DM = 0.30/0.224 = 1.342$, $p = 0.180$
- ▣ 2. factor $\sqrt{(41 - 4 + 2/40)/40} = 0.962$; $HLN = 1.291$, $p = 0.204$ (t_{39} critical value 2.023)
- ▣ 3. i.i.d.: $SE = \sqrt{1.2/40} = 0.173$, $DM = 1.73$: closer to rejection, wrongly, because positive autocorrelation was ignored
- ▣ no evidence that forecast 2 is better

A6: Clark–West by hand [Proposed]

Solution discussed in class

Task

- ▣ A random walk (model 1) is nested in an AR model (model 2); $P = 100$ one-step forecasts: $\overline{e_1^2} = 1.00$, $\overline{e_2^2} = 0.98$, $(\hat{y}_1 - \hat{y}_2)^2 = 0.06$; standard deviations of f_t and of d_t : 0.5. Model: A5.
- ▣ 1. Compute the naive DM statistic for $e_1^2 - e_2^2$ and its one-sided p -value.
- ▣ 2. Compute the Clark–West adjusted mean $\bar{f}$, its t statistic and the one-sided p -value.
- ▣ 3. Explain why the adjustment is added to the larger model, and why DM is undersized here.
- ▣ Report: two statistics, two p -values, two sentences.

A7: Bates–Granger weights [Solved]

Task

- ▣ Two unbiased forecasts with error standard deviations $\sigma_1 = 2$, $\sigma_2 = 3$ and error correlation $\rho = 0.4$.
- ▣ 1. Compute the optimal weight on forecast 1.
- ▣ 2. Compute the combined error variance and compare it with σ_1^2 .
- ▣ 3. Compute the variance with equal weights.
- ▣ Report: one weight, two variances, one sentence.

Solution

- ▣ 1. $\sigma_{12} = 0.4 \cdot 2 \cdot 3 = 2.4$; $D = 4 + 9 - 4.8 = 8.2$;
 $w^* = (9 - 2.4)/8.2 = 0.805$
- ▣ 2. $\sigma_c^2 = (36 - 5.76)/8.2 = 3.688 < 4$: the worse forecast still helps
- ▣ 3. $\frac{1}{4}(4 + 9 + 2 \cdot 2.4) = 4.45$: worse than forecast 1 alone; with very unequal forecasts equal weights can hurt

A8: the cost of estimated weights [Proposed]

Solution discussed in class

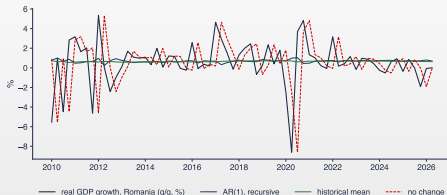
Task

- ▣ $\sigma_1^2 = 1$, $\sigma_2^2 = 1.1$, $\rho = 0.5$; the weight is estimated from $n = 40$ past errors. Model: A7.
- ▣ 1. Show that $\text{MSE}(w) = \text{MSE}(w^*) + (w - w^*)^2 D$ with $D = \sigma_1^2 + \sigma_2^2 - 2\sigma_{12}$.
- ▣ 2. Compute w^* , D and the gain of w^* over $w = 1/2$.
- ▣ 3. A simulation gives $\text{Var}(\hat{w}) = 0.020$ at $n = 40$. Compare the expected estimation loss with the gain and conclude.
- ▣ Report: one identity, three numbers, one conclusion.

B1: forecasting Romanian GDP growth [Solved]

- **Question:** does an AR(1) forecast Romanian quarterly GDP growth better than the historical mean or no change, and does the answer depend on the loss and on 2020?
- **Inputs:** real GDP growth q/q (% , Eurostat, SCA), one-step forecasts made recursively, 2010Q1–2026Q2 ($P = 66$)
- **Tasks**
 1. Compute recursive one-step forecasts from an AR(1), the historical mean and no change, and report RMSE and MAE.
 2. Test AR(1) against each benchmark with DM–HLN under squared and absolute loss.
 3. Repeat the tests without the four quarters of 2020.
 4. Interpretation: is AR(1) a better forecast than the mean, and is it better than no change?
- **Report:** a table of RMSE and MAE, eight HLN statistics with p -values, one sentence
- **Notebook:** section B1

B1: solution [Solved]



Forecast	RMSE	MAE	RMSE, no 2020	MAE, no 2020
AR(1)	2.26	1.48	1.83	1.25
mean	2.24	1.48	1.84	1.26
no change	3.23	2.12	2.78	1.88

- HLN, AR(1) against the mean: 0.43 (p 0.671) squared, -0.12 (p 0.908) absolute; against no change: -1.88 (p 0.065) and -2.58 (p 0.012); without 2020: -2.55 (p 0.013)
- Interpretation: AR(1) and the mean are indistinguishable (the persistence of growth is weak); both clearly beat no change, and the squared-loss evidence is weaker with 2020 because the two COVID quarters dominate the HAC variance

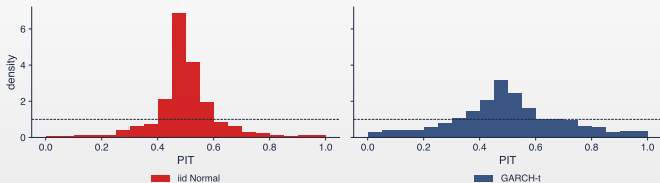
B2: combining SPF unemployment forecasts [Proposed]

- **Question:** can any combination of the individual SPF forecasts of the US unemployment rate beat the simple mean two quarters ahead?
- **Inputs:** SPF variable UNEMP4 (the quarter two quarters after the survey), FRED UNRATE quarterly average, surveys 1990–2025; model: B1 and A7
- **Tasks**
 1. Build the mean, median, 10% trimmed mean, inverse-MSE weights (last 20 surveys with a known outcome, at least 8 forecasts), their 50% shrinkage and the previous best.
 2. Report RMSE relative to the mean and DM–HLN with $h = 3$ against the mean.
 3. Run the Mincer–Zarnowitz test for the mean, with and without 2020.
 4. Interpretation: is the puzzle confirmed for unemployment, and is the mean forecast efficient?
- **Report:** six relative RMSEs, five HLN statistics, two MZ tests, one sentence
- **Notebook:** section B2

B3: density forecasts of EUR/RON [Solved]

- **Question:** is a GARCH(1,1)- t density forecast of daily EUR/RON returns calibrated, and is it better than an i.i.d. Normal forecast?
- **Inputs:** BNR reference rate, $100\Delta \ln S_t$; estimation 2010–2018 (2 270 days), evaluation 2019–2026 (1 925 days), parameters fixed
- **Tasks**
 1. Fit both models by maximum likelihood on the estimation sample and compute one-day PITs on the evaluation sample.
 2. Plot the PIT histograms, run the Berkowitz test and report the share of PITs outside $[0.05, 0.95]$.
 3. Compute the average log score and CRPS and test their differences with DM–HLN.
 4. Interpretation: can the better-scoring model still be miscalibrated, and why?
- **Report:** two histograms, two Berkowitz tests, two shares, four scores, two HLN statistics, one sentence
- **Notebook:** section B3

B3: solution [Solved]



Model	Log score	CRPS	Berkowitz LR	p	Outside 90%
i.i.d. Normal	-0.431	0.0688	1442	< 0.001	1.1%
GARCH(1,1)- t	-1.224	0.0442	539	< 0.001	3.3%

- GARCH- t : $\hat{\alpha} + \hat{\beta} = 0.9999$, $\hat{\nu} = 4.86$; HLN on the score differences: -49.3 (log score), -56.9 (CRPS)
- Interpretation: GARCH- t wins every score by far, yet both are miscalibrated: after 2019 the BNR kept EUR/RON much calmer than in 2010–2018, so densities fixed in 2018 are too wide (a hump, few PITs in the tails); ranking and calibration answer different questions

B4: SPF GDP forecasts and data vintages [Proposed]

- **Question:** does the accuracy and the efficiency of the SPF next-quarter GDP forecast depend on the vintage used as the outcome?
- **Inputs:** SPF mean of $100[(\text{RGDP3}/\text{RGDP2})^4 - 1]$; RTDSM first, second, third release and latest vintage; targets 1990–2025 ($P = 139$); model: B1
- **Tasks**
 1. Compute the RMSE of the SPF mean against each of the four vintages.
 2. Run the Mincer–Zarnowitz test against each vintage.
 3. Recompute the RMSE against the first release and the latest vintage without 2020.
 4. Interpretation: which vintage should a pre-registered evaluation of SPF forecasts use, and why?
- **Report:** four RMSEs, four MZ tests, two RMSEs without 2020, one sentence
- **Notebook:** section B4

B5: load quantiles at the evening peak [Proposed]

- **Question:** are the 90% load intervals for 19:00–20:00 calibrated, and does the expert ARX beat the weekly naive forecast in interval score?
- **Inputs:** Romanian hourly load (Energy-Charts), day-ahead forecasts, 2025–2026 ($P = 638$ days); quantiles from the last 182 errors; model: B3
- **Tasks**
 1. Compute the 5%, 50% and 95% quantile forecasts of the expert ARX and of the weekly naive at the peak hour.
 2. Report the coverage, the shares below and above, the average width and the interval score.
 3. Test the interval-score difference with DM–HLN and report the pinball loss at the median.
 4. Interpretation: is the narrower interval also the better one?
- **Report:** two coverages, four tail shares, two widths, two interval scores, one HLN statistic, one sentence
- **Notebook:** section B5

C1: a pre-registered evaluation by horizon [Proposed]

- **Question:** does a direct AR(3) beat no change for Romanian inflation at some horizon, once we correct for testing four horizons?
- **Inputs:** HICP inflation, y/y ; origins from January 2013, rolling 10-year window; horizons 1, 3, 6, 12 months; model: B1
- **Tasks**
 1. Write the pre-registration: target, horizons, loss, benchmark, test, correction, in five lines, before computing anything.
 2. Compute the relative RMSE and DM-HLN for each horizon.
 3. Apply the Holm correction across the four horizons.
 4. Interpretation: what would you report if you had looked only at the best horizon?
- **Report:** a five-line plan, four relative RMSEs, eight p -values, a project plan
- **Notebook:** section C1

C2: audit an AI answer [Proposed]

- ▣ A student asked an AI assistant to summarise forecast evaluation. The answer:
- ▣ (a) For $h = 4$ use Newey-West with 4 lags; the HLN correction makes the DM test more liberal in small samples.
- ▣ (b) For a point forecast (a point mass) the CRPS equals the absolute error.
- ▣ (c) Expected shortfall is elicitable because it is the mean of the tail, and means are elicitable.
- ▣ (d) To compare an AR model with the random walk it nests, use DM with Normal critical values.
- ▣ (e) A U-shaped PIT histogram means the predictive distribution is too wide.
- ▣ (f) The 90% MCS contains the best model with probability at least 90% asymptotically.
- ▣ (g) With $\sigma_1 = 2$, $\sigma_2 = 3$ and $\rho = 0.4$ the optimal weight on forecast 1 is 0.6.
- ▣ Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number (notebook section C2).
 - ▶ 2. Report: a list of seven verdicts with one line of justification each.

Key takeaways

- ▣ The loss defines the optimal forecast: mean, median, quantile, or a biased value under LinEx
- ▣ Proper scores reward honesty; the log score punishes overconfidence hardest
- ▣ DM needs the HAC variance with $h - 1$ lags and the HLN correction; nested models need Clark–West
- ▣ Estimated weights pay a price of order $1/n$: with similar forecasts, the mean wins
- ▣ An AI answer is a draft: check the lags, the direction of the PIT diagnosis, elicibility and the arithmetic

After the seminar

- Lecture 1 develops each topic of today: loss functions and Bregman losses, calibration and proper scores, DM, GW, Clark–West, SPA and the MCS, combination and real-time data
- Try the [Proposed] tasks in the notebook
- C1 can grow into a team project: combining Romanian inflation forecasts with a pre-registered design
- Reading: Diebold and Mariano (1995); Harvey, Leybourne and Newbold (1997); Gneiting and Raftery (2007); Smith and Wallis (2009)
- **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- Bates, J. M., & Granger, C. W. J. (1969). The combination of forecasts. *Operational Research Quarterly*, 20(4), 451–468. doi:10.1057/jors.1969.103
- Berkowitz, J. (2001). Testing density forecasts, with applications to risk management. *Journal of Business & Economic Statistics*, 19(4), 465–474. doi:10.1198/07350010152596718
- Clark, T. E., & West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1), 291–311. doi:10.1016/j.jeconom.2006.05.023
- Croushore, D., & Stark, T. (2001). A real-time data set for macroeconomists. *Journal of Econometrics*, 105(1), 111–130. doi:10.1016/S0304-4076(01)00072-0
- Diebold, F. X., Gunther, T. A., & Tay, A. S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883. doi:10.2307/2527342
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Fissler, T., & Ziegel, J. F. (2016). Higher order elicibility and Osband's principle. *The Annals of Statistics*, 44(4), 1680–1707. doi:10.1214/16-AOS1439
- Genre, V., Kenny, G., Meyler, A., & Timmermann, A. (2013). Combining expert forecasts: Can anything beat the simple average? *International Journal of Forecasting*, 29(1), 108–121. doi:10.1016/j.ijforecast.2012.06.004
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. doi:10.1198/016214506000001437
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- Harvey, D., Leybourne, S., & Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2), 281–291. doi:10.1016/S0169-2070(96)00719-4
- Mincer, J., & Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (Ed.), *Economic Forecasts and Expectations* (pp. 3–46). NBER. nber.org
- Smith, J., & Wallis, K. F. (2009). A simple explanation of the forecast combination puzzle. *Oxford Bulletin of Economics and Statistics*, 71(3), 331–355. doi:10.1111/j.1468-0084.2008.00541.x